



Date: 2nd August 2026

In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No.: MPBA G506

Course Title: Data Management & Warehousing (DMW)

Credit Units: 3 (2+1)

Lecture: Tuesday & Thursday; 9:00 AM to 9:50 AM; Room no. 6159

Lab: Monday, 11:00 A.M. to 12:50 A.M – P1 batch; Room no. 6117 (IPC LAB)

Wednesday, 11:00 A.M. to 12:50 A.M – P2 batch; Room no. 6117 (IPC LAB)

Instructor-in-Charge: Prof. Abhishek Verma

Instructor: Prof. Abhishek Verma (Till mid-semester exam)

Prof. Debasmita Dey (After mid-semester exam)

1. Course Description

This course introduces students to the practical usage of databases, emphasizing SQL. As big data problems become more prevalent for businesses, this module introduces the basics of databases. After successful completion of the course, students will be able to learn why database management is essential, analyze the data requirements of a business scenario and represent these requirements utilizing entity-relationship (ER) diagrams, translate an ER diagram into normalized tables for a relational database management system, and write simple and relatively complex data retrieval queries. Students will learn how to extract valuable insights from databases through data analytics powered by SQL.

2. Scope and Objectives

The course covers various Database Management and Data Warehousing concepts and enhances learning effectiveness through developing scripting skills. It aims to provide students with the necessary knowledge and skills to understand the data infrastructure underpinning business intelligence and analytics. Students undertaking this course should be able to appreciate the complexities involved in Database management and be prepared to solve the issues a business analyst faces.

3. Course learning outcomes:

- Understand the database environment and the complete development process, including requirements gathering, design, and implementation.
- Model data using Entity-Relationship diagrams and design logical and physical relational databases.
- Write effective SQL queries to manage and manipulate data within relational databases.
- Explain the architecture of database applications, foundations of data warehousing, and principles of data quality and integration.
- Explore the role of databases in Big Data analytics and understand essential tasks in database administration.

4. Text Book

TB1. Database System Concepts by *Abraham Silberschatz, Henry F. Korth, S. Sudarshan*

5. Reference Books

RB1. Fundamentals of Database Systems by *Elmasri Ramez and Navathe Shamkant* (2017) 7th Ed. Pearson (ISBN: 9789332582705).

RB2. Graph Databases Specs By *Ian Robinson, Jim Webber, and Emil Eifrem* Publisher: O'Reilly Media





6. Theory class modules

Module No.	Topic	Learning Outcome	Reference/ Reading
1	Introduction to Database	Purpose of Database Systems, View of Data, Database Languages, Database Design, Database Engine, Database and Application Architecture, History of Database Systems	TB1 -CH 1
2	Relational Model	Structure of Relational Databases, Database Schema, Keys, Schema Diagrams, Relational Query Languages	TB1 -CH 2
3	Database Design Using the E-R Model	Overview of the Design Process, The Entity-Relationship Model, Primary Key, Removing Redundant Attributes in Entity Sets, Reducing E-R Diagrams to Relational Schemas, Entity-Relationship Design Issues	TB1 -CH 6
4	Relational Database Design	Features of Good Relational Designs Decomposition Using Functional Dependencies, Normal Forms, Functional-Dependency Theory, Algorithms for Decomposition Using Functional Dependencies, Decomposition Using Multivalued Dependencies, More Normal Forms	TB1 -CH 7
5	Structured Query Language	SQL constructs and syntax for data handling	TB1 -CH 17
6	Complex Data Types	Semi-structured Data, Textual Data	TB1 -CH 3, 4
7	Data and database administration	Functions of database administration, changing role of database users	
Mid-Term Examination			
8	Transactions	PL SQL, Transaction Concurrency, ACID Property · Transaction States & Schedule · Need for Concurrency Control	TB1 -CH 17
9	Serializability	· Serializability, Precedence Graph, View Serializability	TB1 -CH 17
10	Concurrency Control	· Locking Protocol – Shared Exclusive, 2 Phase Locking · Timestamp ordering · Recovery system · Failure types	TB1 – CH 18,19
11	Storage Management	· Physical Storage Systems - RAID · Storage Architecture- Data-Dictionary Storage, Buffer, Column-Oriented Storage · Indexing, Types of Indexing	TB1 – CH 12,13,14
12	Data Warehousing and Analytics	· Data Warehouse- Concept, characteristics, OLTP vs OLAP, ETL, ELT · Schema · Data Transform, pre-processing, and cleaning · Data mart and Data Lake	TB1 – CH 11
13	Big Data Analytics	· Small data vs Big data · Hadoop framework, working principle, benefits and challenges · MongoDB · Spark, Pyspark · Latest trends- Vector database, Serverless & Cloud-Native DB	TB1 – CH 10
14	Data Mining	Information Retrieval and pattern recognition from database	TB1 – CH 11





Practical lab modules

Module#	Topic	Learning outcome
1	Introduction to databases	Importance and application of DBMS
2	Preparing E-R diagrams	Data modeling with the entity-relationship (E-R) diagrams
3	Preparing a relational model using logical database design	Apply normalization
4	Data Definition in SQL	Data definition queries
5	Data Manipulation in SQL	Single table queries
6	Data Manipulation in SQL	Creating and using views
7	Data Manipulation in SQL	Various joins in SQL
8	Data Manipulation in SQL	Subqueries and Unions
Mid-Term Examination		
9	Transaction Management	Transaction handling and management
10	Data Warehouse and Analytics	OLAP, ETL, ELT, Data pre-processing, and cleaning
11	Graph databases	NoSQL and other semi-structured databases
12	Big Data	PySpark

7. Pedagogy

Teaching will consist of Lectures and Labs. The assessments will be conducted online and offline, and 10% of the total marks (class participation section) will be conducted as surprise quizzes in viva format. The total marks for the course are 200.

8. Evaluation scheme

	Evaluation component	Weightage (%)	Date & Time	Nature of component
Pre-Mid Components	Pre mid-sem Quizzes	10%	TBA in Class	Open Book
	Assignments	10%	TBA in Class	Open Book
	Lab tests	10%	TBA in Class	Open Book
	Mid-term exam	20%	9 th October 2026 4:00 PM-5:30 PM	Closed book
After Mid-Sem				
Post-Mid Components	In-class/lab Assignment	10%	Weekly once	Open book/notes/ Internet
	Post-MID Sem Quiz	10%	1st Week of November	TBA
	Comprehensive Exam	30%	14th December 2026 2.00 PM-5.00 PM	Closed book

9. Chamber Consultation Hour:

- Prof. Abhishek Verma [Pre-Mid Classes] – Every Tuesday, 3:30 P.M. - 4:30 P.M.
- Prof. Debasmitta Dey [Post-Mid Classes]- TBA/ via prior appointment only.

Online appointments can be booked for time slots other than chamber consultation hours (subject to availability)





10. Notes

- The highest level of intellectual integrity is expected of students while they work on lab tests/quizzes in terms of cheating & plagiarism. Students are also expected to submit their assignments on time, failing which these will not be evaluated. No exceptions will be entertained for the acceptance of late submissions.
- No make-up would be allowed for any open-book components and Mid-Semester exam. For other components proper medical certification from the **institute medical center** clearly specifying you are unfit to write the test will be essential.

Instructor-in-charge
Course No.: MPBA G506

